

Paper Id. 250745

Printed Page: 1 of 2
Subject Code: RCA1601

Roll No:

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MCA-INT
(SEM VI) THEORY EXAMINATION 2024-25
DATABASE MANAGEMENT SYSTEMS

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A1. Attempt *all* questions in brief.

02 x 7 = 14

Q no.	Question	CO	Level
a.	Define a weak entity set with an example.	1	K1
b.	What is a candidate key? How is it different from a primary key?	2	K2
c.	List any two advantages of relational databases.	1	K1
d.	What is a tuple and an attribute in the context of relational databases?	2	K1
e.	Define a trigger in SQL.	3	K1
f.	Explain B-tree indexing.	4	K2
g.	What is a deadlock in DBMS?	5	K1

SECTION B2. Attempt any *three* of the following:

07 x 3 = 21

Q no.	Question	CO	Level
a.	Justify with assumptions to draw ER Diagram of University Examination System which demonstrates the following Entities with single value Attributes and Multi-valued attribute and all possible cardinalities. Entities: University, Student, Course, Department	1	K5
b.	Explain three levels of abstraction or schemas architecture of DBMS	2	K1
c.	Analyze the importance of following terms: (i) Relational data model concepts (ii) Mapping Constraints	2	K4
d.	Explain cost-based query optimization with example.	4	K4
e.	What are the common strategies used to prevent deadlocks in DBMS?	5	K1

SECTION C3. Attempt any *one* part of the following:

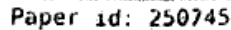
07 x 1 = 07

Q no.	Question	CO	Level
a.	Explain the different types of relationships in E-R modeling with suitable examples.	1	K2
b.	Discuss the role of DBMS in an enterprise with a real-world example.	1	K5

4. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	Write and explain any five relational algebra operations with examples.	2	K2
b.	Define functional dependency. Explain Armstrong's axioms with examples.	2	K2



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$$\underline{07} \times 1 = 07$$
$$\underline{07} \times \underline{1} = \underline{07}$$
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